Karan Rajput

Rajsamand, India +91-8824806030 rajputkaran997@gmail.com
leetcode.com linkedin.com/in/karan-rajput github.com

Profile Summary

Integrated M.Tech student in Computational Data Science with strong foundations in **machine learning**, and **data analysis**. Experienced in building end-to-end ML pipelines for NLP-based projects and passionate about solving real-world problems using data-driven approaches.

Education

- **Vellore Institute of Technology** (Oct 2022 – 2027)
  Integrated M.Tech in Computer Science and Engineering *CGPA: 8.59/10*

Technical Skills

- **Programming:** Java , Python
- **Core CS:** Object-Oriented Programming, Data Structures(Arrays, Strings, Hashing, Stack, Queue)
- **Machine Learning & NLP:** Scikit-learn, Text Classification, Model Evaluation
- **Data Analysis:** Pandas, NumPy, Matplotlib, Seaborn
- **Tools:** GitHub, VS Code, Google Colab

Projects

AI-Driven News Credibility and Influence Analysis *Jan 2026 – Present*
Python, NLP, Scikit-learn, HuggingFace, Pandas

- Building an end-to-end NLP pipeline to classify news articles as credible or non-credible.
- Processed a dataset of approximately 45,000 labeled news articles.
- Split the dataset into training (∼40K samples) and testing (∼4.5K samples).
- Applied text preprocessing and TF-IDF vectorization resulting in high-dimensional feature representations.
- Trained and evaluated multiple supervised machine learning models.
- Assessed model performance using accuracy, precision, recall, and F1-score.

Fake News Detection System *Dec 2025*
Python, NLP, Scikit-learn

- Developed a supervised machine learning model to detect fake news from textual data.
- Applied NLP preprocessing techniques such as tokenization, stopword removal, and vectorization.
- Experimented with multiple classification algorithms and performed hyperparameter tuning.
- Analyzed model performance using standard evaluation metrics.

Sentiment Analysis on Social Media Data *Nov 2025*
Python, NLP, Scikit-learn, Pandas

- Built a sentiment analysis system to classify user opinions as positive, negative, or neutral.
- Preprocessed raw text data using cleaning, tokenization, and vectorization techniques.
- Trained and evaluated supervised learning models to analyze sentiment trends.
- Gained practical experience in handling noisy real-world text data.

Student Performance Analysis *Oct 2025*
Python, Pandas, NumPy, Matplotlib, Seaborn

- Performed data analysis on student academic performance data to identify patterns and trends.
- Conducted exploratory data analysis (EDA) using Pandas to understand distributions and relationships.
- Visualized key insights using Matplotlib and Seaborn to highlight factors influencing student performance.
- Interpreted analytical results to support data-driven decision making.

Certifications

- Introduction to Machine Learning – NPTEL (IIT Madras) *Oct 2024*
- Python for Data Science – Coursera *2024*

Soft Skills & Interests

- **Soft Skills:** Problem Solving, Analytical Thinking, Team Collaboration
- **Languages:** English, Hindi